Assignment 6. Midpint and Trapezoidal method Marks 20

Posted on 10.09.2026 @ 2:30 pm and due on 10.09.2026 @ 5:40 pm

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1. Write library functions for Midpoint and Trapezoidal numerical methods for integration. **[10]**

2. Using both Midpoint and Trapezoidal method evaluate the following integrals and compare with the analytical results, for $N = 4$, 8, 15, 20. Present the result in tabular form. **[10]**

$$\int_1^2 \frac{dx}{x} = 0.69314718, \qquad \int_0^{\pi/2} x\,\cos x\,dx = \frac{\pi}{2} - 1, \qquad \int_0^1 x\,\tan^{-1} x\,dx = \frac{\pi}{4} - \frac{1}{2}$$